KARTHIK MANNEM

Los Angeles, CA | 747-347-6409 | mannemkarthik01@gmail.com
linkedin.com/in/karthik-mannem-2008b4225 | github.com/mannemkarthik010 | karthikmannem.com

Technical Skills

Languages: Python, SQL, Java, TypeScript
ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, Gradient Boosting, Time-Series Forecasting, Backtesting
NLP & LLM: Transformers, HuggingFace, LangChain, RAG, Tool-Calling Agents, Prompt Engineering, Eval Harnesses
Data: Pandas, NumPy, Feature Engineering, PostgreSQL, MySQL, MongoDB, ChromaDB / Vector Search
MLOps & Cloud: AWS (SageMaker, Bedrock, Lambda, API Gateway, S3), Terraform, Docker, GitHub Actions, Flask

Experience

•AI Engineer
SLN Solutions Oct 2026 – Present

– Building ML models to forecast demand and flag stock-out risk from retail sales and stock-transaction data.
– Serving shortage-risk scores and alerts to the client's inventory app via Python REST APIs and SQL pipelines.

•AI Model Evaluation Analyst *Remote*
Handshake AI Sep 2025 – Apr 2026

– Scored 1,000+ LLM responses against rubrics for factual accuracy, reasoning quality, and instruction-following.
– Catalogued recurring failure and hallucination patterns into reusable categories that fed prompt refinement.

•Machine Learning Engineer (Freelance) *Remote*
Self-Employed Aug 2023 – Jun 2024

– Delivered deep learning models (LSTM, CNN) for 3 client applications, owning preprocessing, cross-validated training, and a Dockerized Flask inference API.
– Cut training time by roughly 25% through pipeline profiling, batched preprocessing, and training-config tuning.

Projects

•Retail Demand Intelligence Platform – Forecasting, Anomaly Detection & Agentic Q&A *GitHub*
Python, scikit-learn, Flask, Anthropic/OpenAI/Bedrock, Terraform, AWS, Docker, GitHub Actions Live Demo

– Built a 3-model pipeline (seasonal-naive, Holt-Winters, GBM) over 240 series, rolling-origin backtested, selecting a winner per series (GBM: 7.9% WAPE, best on 154/240).
– Shipped 80% prediction intervals and occlusion-based feature attribution, reporting measured coverage on held-out data (74.2% vs. 80% nominal) instead of asserting calibration.
– Designed a 6-tool agentic Q&A layer (pluggable LLM backend, audit logging, 37-case routing eval on a mock router in CI) and a drift-detection loop that re-selects models per series; AWS stack in Terraform, tested on mocks.

•TaxProof – Evidence-Linked Tax Workpaper, Trustworthy-AI Prototype *GitHub*
Python, FastAPI, PostgreSQL, Next.js, TypeScript, Pytest, Vitest, Vercel Live Demo

– Built an evidence graph over a SHA-256-frozen sample tax packet that traces each return line through its rule to the source page, plus a challenge engine that flags contradictions and misreads.
– Enforced roles server-side with masked sensitive fields and a hash-chained audit log that detects any edited or deleted row; constrained the AI to cited evidence, so it investigates but never calculates.
– Shipped in 10 tested iterations: 230 backend tests (SQLite and PostgreSQL) and 36 frontend tests; live on Vercel.

•Preventive Care Recommendation Agent – Agentic RAG, Educational Prototype *GitHub*
Python, LangGraph, LangChain, ChromaDB, Llama-3.1, Streamlit Live Demo

– Built a LangGraph agent that retrieves passages from a ChromaDB guideline index, drafts a structured care plan grounded in them, and lists the retrieved sources.
– Implemented the ACC/AHA Pooled Cohort Equations in Python for 10-year ASCVD risk, verified on the guideline's worked example, plus a critic loop (max 2 revisions).

Open-Source Contributions

unionai-oss/pandera – 1 PR merged: #2437 itemsize-parameterized NumPy dtypes (`S16`, `U16`), with tests.
pytorch/pytorch – 1 PR approved, open: #191687 fixes `torch.fx` codegen for escaped names, with tests.

Education

•California State University, Northridge *Los Angeles, CA*
Master of Science in Computer Science | ***CGPA: 3.92 / 4.0*** Aug 2024 – May 2026

•Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology *Chennai, India*
Bachelor of Technology in Computer Science and Engineering | ***CGPA: 9.21 / 10.00*** Aug 2020 – May 2024

Leadership & Certifications

Board of Directors, Facilities & Ops – USU, CSUN (2024–26): 38,000+ students, $5M budget.
Certifications: AWS AI & ML (Foundation, Associate) · NVIDIA Deep Learning Fundamentals